

División larga de polinomios (4)

$$(1) \frac{x^3 - 8}{x - 2}.$$

Complete con términos faltantes: $x^3 + 0x^2 + 0x - 8$.

$$x^3 \div x = x^2 \Rightarrow (x - 2) \cdot x^2 = x^3 - 2x^2.$$

$$\underline{(x^3 + 0x^2) - (x^3 - 2x^2)} = 2x^2.$$

$$2x^2 \div x = 2x \Rightarrow (x - 2) \cdot 2x = 2x^2 - 4x.$$

$$\underline{(2x^2 + 0x) - (2x^2 - 4x)} = 4x.$$

$$4x \div x = 4 \Rightarrow (x - 2) \cdot 4 = 4x - 8.$$

$$\underline{(4x - 8) - (4x - 8)} = 0.$$

$$\boxed{\text{Cociente} = x^2 + 2x + 4, \quad \text{Residuo} = 0}.$$

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4).$$

$$(2) \frac{2x^4 - 3x^3 + 0x^2 + 5x - 6}{x^2 - x + 2}.$$

$$2x^4 \div x^2 = 2x^2 \Rightarrow (x^2 - x + 2)(2x^2) = 2x^4 - 2x^3 + 4x^2.$$

$$\underline{(2x^4 - 3x^3 + 0x^2) - (2x^4 - 2x^3 + 4x^2)} = -x^3 - 4x^2.$$

$$(-x^3) \div x^2 = -x \Rightarrow (x^2 - x + 2)(-x) = -x^3 + x^2 - 2x.$$

$$\underline{(-x^3 - 4x^2 + 5x) - (-x^3 + x^2 - 2x)} = -5x^2 + 7x.$$

$$(-5x^2) \div x^2 = -5 \Rightarrow (x^2 - x + 2)(-5) = -5x^2 + 5x - 10.$$

$$\underline{(-5x^2 + 7x - 6) - (-5x^2 + 5x - 10)} = 2x + 4.$$

$$\boxed{\text{Cociente} = 2x^2 - x - 5, \quad \text{Residuo} = 2x + 4}.$$

$$2x^4 - 3x^3 + 5x - 6 = (x^2 - x + 2)(2x^2 - x - 5) + (2x + 4).$$

$$(3) \frac{4x^5 + 0x^4 + x^2 - 7}{2x^2 + 1}.$$

$$4x^5 \div 2x^2 = 2x^3 \Rightarrow (2x^2 + 1)(2x^3) = 4x^5 + 2x^3.$$

$$\underline{(4x^5 + 0x^4 + 0x^3) - (4x^5 + 2x^3) = -2x^3.}$$

$$(-2x^3) \div 2x^2 = -x \Rightarrow (2x^2 + 1)(-x) = -2x^3 - x.$$

$$\underline{(-2x^3 + 0x^2 + 0x) - (-2x^3 - x) = x.}$$

$$x \div 2x^2 = \frac{1}{2x} \text{ (no procede en polinomios)} \Rightarrow \text{bajamos } +x^2 - 7.$$

$$\text{Trabajamos con el término siguiente: } x^2 \div 2x^2 = \frac{1}{2}.$$

$$(2x^2 + 1)\left(\frac{1}{2}\right) = x^2 + \frac{1}{2}.$$

$$\underline{(x^2 + x - 7) - (x^2 + \frac{1}{2}) = x - \frac{15}{2}.}$$

$\text{Cociente} = 2x^3 - x + \frac{1}{2}, \quad \text{Residuo} = x - \frac{15}{2}.$
--

$$4x^5 + x^2 - 7 = (2x^2 + 1)\left(2x^3 - x + \frac{1}{2}\right) + \left(x - \frac{15}{2}\right).$$

$$(4) \frac{-x^4 + 6x^2 - 9x + 4}{-x^2 + 3x - 2}.$$

$$(-x^4) \div (-x^2) = x^2 \Rightarrow (-x^2 + 3x - 2)(x^2) = -x^4 + 3x^3 - 2x^2.$$

$$\underline{(-x^4 + 0x^3 + 6x^2) - (-x^4 + 3x^3 - 2x^2) = -3x^3 + 8x^2.}$$

$$(-3x^3) \div (-x^2) = 3x \Rightarrow (-x^2 + 3x - 2)(3x) = -3x^3 + 9x^2 - 6x.$$

$$\underline{(-3x^3 + 8x^2 - 9x) - (-3x^3 + 9x^2 - 6x) = -x^2 - 3x.}$$

$$(-x^2) \div (-x^2) = 1 \Rightarrow (-x^2 + 3x - 2)(1) = -x^2 + 3x - 2.$$

$$\underline{(-x^2 - 3x + 4) - (-x^2 + 3x - 2) = -6x + 6.}$$

$\text{Cociente} = x^2 + 3x + 1, \quad \text{Residuo} = -6x + 6.$

$$-x^4 + 6x^2 - 9x + 4 = (-x^2 + 3x - 2)(x^2 + 3x + 1) + (-6x + 6).$$

División sintética (4) — divisor lineal $x - a$

Convención: fila 1 = coeficientes; fila 2 = productos acumulados; fila 3 = sumas. La última entrada es el residuo.

(5) $P(x) = 3x^4 - 2x^3 + 0x^2 + 5x - 1$, divisor $x - 2$ ($a = 2$).

$$\begin{array}{r|rrrrrr} 2 & 3 & -2 & 0 & 5 & -1 \\ & & 6 & 8 & 16 & 42 \\ \hline & 3 & 4 & 8 & 21 & \boxed{41} \end{array}$$

$$\boxed{Q(x) = 3x^3 + 4x^2 + 8x + 21, \quad R = 41}, y$$

$$P(x) = (x - 2) Q(x) + R.$$

(6) $P(x) = x^5 - 4x^4 + 0x^3 + x^2 + 0x - 6$, divisor $x + 1$ ($a = -1$).

$$\begin{array}{r|rrrrrr} -1 & 1 & -4 & 0 & 1 & 0 & -6 \\ & & -1 & 5 & -5 & 4 & -4 \\ \hline & 1 & -5 & 5 & -4 & 4 & \boxed{-10} \end{array}$$

$$\boxed{Q(x) = x^4 - 5x^3 + 5x^2 - 4x + 4, \quad R = -10}.$$

(7) $P(x) = 2x^3 - 5x^2 + 2x + 3$, divisor $x - 3$ ($a = 3$).

$$\begin{array}{r|rrrr} 3 & 2 & -5 & 2 & 3 \\ & & 6 & 3 & 15 \\ \hline & 2 & 1 & 5 & \boxed{18} \end{array}$$

$$\boxed{Q(x) = 2x^2 + x + 5, \quad R = 18}.$$

(8) $P(x) = 4x^4 + 0x^3 + 0x^2 - 5x + 2$, divisor $x - \frac{1}{2}$ ($a = \frac{1}{2}$).

$$\begin{array}{r|rrrrr} \frac{1}{2} & 4 & 0 & 0 & -5 & 2 \\ & & 2 & 1 & \frac{1}{2} & -\frac{9}{4} \\ \hline & 4 & 2 & 1 & -\frac{9}{2} & \boxed{-\frac{1}{4}} \end{array}$$

$$\boxed{Q(x) = 4x^3 + 2x^2 + x - \frac{9}{2}, \quad R = -\frac{1}{4}}.$$

Teorema del residuo (2)

(9) $f(x) = 2x^4 - 5x^3 + 3x - 7$, residuo al dividir entre $x + 3$ ($a = -3$):

$$\begin{aligned} f(-3) &= 2(-3)^4 - 5(-3)^3 + 3(-3) - 7 \\ &= 2(81) - 5(-27) - 9 - 7 \\ &= 162 + 135 - 16 \\ &= \boxed{281}. \end{aligned}$$

(10) $f(x) = -x^5 + 4x^2 - 1$, residuo al dividir entre $x - 2$ ($a = 2$):

$$\begin{aligned} f(2) &= -(2)^5 + 4(2)^2 - 1 \\ &= -32 + 16 - 1 \\ &= \boxed{-17}. \end{aligned}$$

Teorema del factor (2)

(11) Verificar si $x - 1$ es factor de $f(x) = x^3 - 6x^2 + 11x - 6$.

$$f(1) = 1 - 6 + 11 - 6 = 0 \Rightarrow \boxed{x - 1 \text{ es factor}}.$$

División sintética con $a = 1$:

$$\begin{array}{r|rrrr} 1 & 1 & -6 & 11 & -6 \\ & & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

$$\Rightarrow Q(x) = x^2 - 5x + 6 = (x - 2)(x - 3).$$

$$\boxed{f(x) = (x - 1)(x - 2)(x - 3)}.$$

(12) Verificar si $x + 2$ es factor de $f(x) = 2x^3 + 3x^2 - 8x - 12$.

$$f(-2) = 2(-8) + 3(4) - 8(-2) - 12 = 0 \Rightarrow \boxed{x + 2 \text{ es factor}}.$$

División sintética con $a = -2$:

$$\begin{array}{r|rrrr} -2 & 2 & 3 & -8 & -12 \\ & & -4 & 2 & 12 \\ \hline & 2 & -1 & -6 & 0 \end{array}$$

$$\Rightarrow Q(x) = 2x^2 - x - 6 = (2x + 3)(x - 2).$$

$$\boxed{f(x) = (x + 2)(x - 2)(2x + 3)}.$$